CSE 202: Object Oriented Programming — Unit 3

File Handling, Constructors & Destructors (Shuffled Order)

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1. Which stream class is used exclusively for reading data from a file?

A. `ofstream`

B. `ifstream`

C. `fstream`

D. `iostream`

> **Correct Answer:** B. `ifstream`
> *Explanation:* `ifstream` stands for Input File Stream and is meant solely for reading data.

2. Which header file is required for file handling classes (`ifstream`, `ofstream`, `fstream`) in C++?

A. ``

B. ``

C. ``

D. ``

> **Correct Answer:** B. ``

3. What does the file mode `ios::app` guarantee during write operations?

A. It truncates the file content before writing

B. It opens the file for reading only

C. Every write operation goes to the end of the file

D. It positions the pointer at the beginning

Correct Answer: C. Every write operation goes to the end of the file

4. Which stream class is used exclusively for writing data to a file?

A. `ifstream`

B. `fstream`

C. `ofstream`

D. `iostream`

Correct Answer: C. `ofstream`

5. Which file mode deletes/truncates existing contents of a file as soon as it is opened?

A. `ios::trunc`

B. `ios::app`

C. `ios::ate`

D. `ios::binary`

Correct Answer: A. `ios::trunc`

6. Which stream class supports both reading from and writing to the same file?

A. `ifstream`

B. `ofstream`

C. `fstream`

D. `stdio`

Correct Answer: C. `fstream`

7. What does `ios::ate` do differently compared to `ios::app` when opening a file?

A. It forces all subsequent writes to occur strictly at the end

B. It positions the file pointer at the end initially, but allows later writes anywhere

C. It discards existing file content immediately

D. It opens the file exclusively in binary mode

Correct Answer: B. It positions the file pointer at the end initially, but allows later writes anywhere

8. Which function is used to explicitly associate a stream object with a file?

A. `open()`

B. `attach()`

C. `bind()`

D. `associate()`

Correct Answer: A. `open()`

9. What is performed by the code snippet below?

```
ofstream fout("data.txt", ios::out);
fout << "Hello";
```

A. Reads "Hello" from data.txt

B. Writes "Hello" into data.txt

C. Checks if data.txt exists

D. Appends "Hello" at the end of data.txt

Correct Answer: B. Writes "Hello" into data.txt

10. Which function should be called after finishing file operations to flush buffers and release system resources?

A. `flush()`

B. `close()`

C. `clear()`

D. `release()`

Correct Answer: B. `close()`

11. Which function is commonly used to read an entire line including spaces from a file stream?

A. `get()`

B. `getline()`

C. `read()`

D. `scanf()`

Correct Answer: B. `getline()`

12. Which file mode opens a file so data is read/written raw as stored in memory without newline translations?

A. `ios::in`

B. `ios::binary`

C. `ios::app`

D. `ios::nocreate`

Correct Answer: B. `ios::binary`

13. Which stream member function tests whether the end-of-file condition has been reached?

A. `good()`

B. `eof()`

C. `fail()`

D. `bad()`

Correct Answer: B. `eof()`

14. Which pair of stream functions is typically used to read and write raw binary data of a specified byte count?

A. `get()` and `put()`

B. `read()` and `write()`

C. `input()` and `output()`

D. `seek()` and `tell()`

Correct Answer: B. `read()` and `write()`

15. What operation is performed by `file.read((char*)&obj, sizeof(obj));`?

A. Writing formatted text data into an object

B. Reading binary record bytes directly into an object's memory

C. Calculating the size of an object

D. Deleting an object from the file

Correct Answer: B. Reading binary record bytes directly into an object's memory

16. In sequential file access, how are records accessed?

A. Direct jump to any record via index

B. One after another in the order they are stored

C. Random order based on memory address

D. Reverse order only

Correct Answer: B. One after another in the order they are stored

17. Which functions are used to reposition the read (get) and write (put) file pointers for random file access?

A. `tellg()` and `tellp()`

B. `moveg()` and `movep()`

C. `seekg()` and `seekp()`

D. `posg()` and `posp()`

Correct Answer: C. `seekg()` and `seekp()`

18. Which function returns the current position of the read (get) pointer in a file stream?

A. `tellg()`

B. `tellp()`

C. `seekg()`

D. `getpos()`

Correct Answer: A. `tellg()`

19. What does the expression `fout.seekp(0, ios::beg);` do?

A. Moves the write pointer to the beginning of the file

B. Moves the read pointer to the end of the file

C. Deletes the first character in the file

D. Returns current write position

Correct Answer: A. Moves the write pointer to the beginning of the file

20. In random access file processing, how is the exact byte location of a specific record calculated?

A. `record_number + sizeof(record)`

B. `record_number * sizeof(record)`

C. `sizeof(record) / record_number`

D. `tellg() + seekg()`

Correct Answer: B. `record_number * sizeof(record)`

21. What actually gets stored in a binary file when writing a class object using `write()`?

A. The class declaration code

B. Raw memory bytes of the object's data layout

C. Function names and variable types

D. XML string representation

Correct Answer: B. Raw memory bytes of the object's data layout

22. What parameter casting is required when calling `write()` or `read()` for a custom object?

A. `(int*)&obj`

B. `(char*)&obj`

C. `(void*)&obj`

D. `(string*)&obj`

Correct Answer: B. `(char*)&obj`

23. Why can C++ `struct` instances be written to/read from files using the same approach as `class` instances?

A. `struct` cannot have data members

B. In C++, `struct` is a user-defined type with data members identical to `class` except default access level

C. `struct` is automatically converted to string

D. `struct` memory cannot be accessed directly

Correct Answer: B. In C++, `struct` is a user-defined type with data members identical to `class` except default access level

24. What is a constructor in C++?

A. A special static function used to delete objects

B. A special member function automatically invoked when an object is created

C. A built-in loop construct

D. A operator used to overload addition

Correct Answer: B. A special member function automatically invoked when an object is created

25. What defines a Default Constructor in C++?

A. A constructor that requires at least two integer parameters

B. A constructor that accepts no arguments or where all arguments have default values

C. A constructor that returns an integer code

D. A constructor defined outside the class

Correct Answer: B. A constructor that accepts no arguments or where all arguments have default values

26. What type of constructor is shown in the code below?

```
class Box {
public:
    int length;
    Box(int l = 10) {
        length = l;
    }
};
```

A. Copy constructor

B. Constructor with default argument

C. Virtual constructor

D. Destructor

Correct Answer: B. Constructor with default argument

27. What is the primary purpose of a destructor in C++?

A. To initialize data members when an object is allocated

B. To perform cleanup/resource release when an object goes out of scope or is deleted

C. To copy one object to another

D. To overload operators

Correct Answer: B. To perform cleanup/resource release when an object goes out of scope or is deleted

28. Why are Member Initializer Lists primarily used in C++ constructors?

A. To initialize `const` members, reference members, and base class members before constructor body runs

B. To speed up `cout` output

C. To make private members public

D. To avoid using header files

Correct Answer: A. To initialize `const` members, reference members, and base class members before constructor body runs

29. What is a Parameterized Constructor?

A. A constructor that accepts arguments to initialize an object with specific values

B. A constructor with no arguments

C. A constructor that deletes objects

D. A constructor that cannot be overloaded

Correct Answer: A. A constructor that accepts arguments to initialize an object with specific values

30. What kind of constructor is demonstrated in the code below?

```
class Point {
public:
    int x, y;
    Point(const Point &p) {
        x = p.x;
        y = p.y;
    }
};
```

A. Default constructor

B. Copy constructor

C. Move constructor

D. Parameterized constructor

Correct Answer: B. Copy constructor
Explanation: A copy constructor accepts a reference to an existing object of the same class (`const Point &p`) to construct a new copy.